

ALAN M VARGHESE

LinkedIn | GitHub | Portfolio
DOB – 05/06/2005

alanmvarghese.mec@gmail.com
+91 8594048542

SKILLS & INTERESTS

- Technical Skills:** HTML, CSS, JavaScript, C, C++, C#, Python, MySQL, Java, R, AWS S3, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Tableau, Computer Graphics, Unity, Prototyping, Scripting, UI/UX Design, Git/Github
- Soft Skills:** Communication, Problem Solving, Teamwork, Leadership, Time Management, Interpersonal Skills
- Interests:** Web Development, Cyber Security, Digital Marketing

EDUCATION

- Govt. Model Engineering College** **7.59 | 2027**
KTU, B.Tech in Computer Science and Business Systems
- St. Vincent EMHSS, Pala State, 12th** **95.5% | 2023**
- Carmel Residential Senior Secondary School** **90.2% | 2021**
CBSE, 10th

WORK EXPERIENCE

- Beehive.in** **4 Months**
Software Developer
Technologies Used: HTML, CSS, JavaScript, Python, React, Flask, Node.js, MySQL, Next.js, Prisma, Express.js, AWS S3, Tableau
During my internship I managed a portfolio of more than 15 clients and accelerate digital transformation through high-performance web ecosystems. I architected and engineered a proprietary, full-stack e-commerce software that enables businesses to instantly launch and scale their stores online. The platform empowers merchants with a robust admin panel for seamless product uploading and inventory management, paired with a dynamic admin dashboard providing real-time store analytics. Through these scalable ecosystems, I have successfully deployed commercial platforms that consistently generate over 100 sales monthly.
- Laneway India** **1 Month**
WordPress Developer
Technologies Used: WordPress, Elementor, YoastSEO, Rankmath SEO
During my front-end developer internship at Laneway, I led the UI/UX design and development of more than 3 responsive web projects using WordPress and HTML. In addition to technical deployment, I spearheaded the intern onboarding program, designing and teaching comprehensive WordPress development classes to a cohort of around 30 new interns, successfully accelerating their time-to-productivity.

PROJECTS

- JUNGLE DASH - Browser-Based Endless Runner Game** **Team Size: 1**
Gameplay Systems & Mechanics Programmer **1 Months**
Technologies Used: HTML5 Canvas/DOM, CSS3, JavaScript (ES6+), Web APIs, LocalStorage
Engineered a responsive, multi-platform 3-lane endless runner utilizing vanilla JavaScript and web technologies. Designed an extensible architecture to manage 8 unique structural obstacle variations. Built custom gameplay state machines including a progressive speed-scaling engine, an invincibility-frame loop, and an immediate-response pausing framework. Optimised cross-platform accessibility by implementing unified touch-swipe and keyboard input listener layers, utilizing LocalStorage for low-overhead local profile high-score persistence.
- DEEP SEA DESCENT - Browser-Based Endless Ocean Exploration Game** **Team Size: 1**
Graphics & Gameplay State Programmer **1 Months**
Technologies Used: HTML5 Canvas/Web APIs, CSS3, JavaScript (ES6+), LocalStorage
Engineered a responsive, multi-platform 2D underwater survival game featuring a multi-tier dynamic game loop that algorithms scale across 5 distinct depth-based environmental zones. Programmed custom procedural animation scripts for 5 marine creature classes utilizing coordinate transforms and mathematical sinusoidal wave functions to simulate organic movement patterns. Architected a complex dual-resource survival game tracking structural hull integrity alongside a non-linear atmospheric pressure gauge, while developing a high-fidelity visual feedback pipeline integrating real-time Canvas screen-shake algorithms, state-driven red vignette overlays, and low-latency, unified touch and keyboard input layers.
- QUISHY** **Team Size: 4**
Software Developer **3 Months**
Technologies Used: HTML, CSS, Python, Flask, Random Forest, Scikit-learn, Pandas, Regex
Engineered a specialized cybersecurity solution combining Machine Learning and heuristic analysis to detect advanced QR code phishing (quishing) vectors. Architected a Python and Flask-based system that processes and decodes QR images to isolate embedded URLs, instantly subjecting them to structural analysis via regex-driven keyword detection and Shannon entropy computation to flag obfuscated links. To deliver robust predictive defense, trained and deployed a Random Forest classifier using Scikit-learn and Pandas, achieving high-accuracy detection of malicious URLs before user exposure, all served through a clean HTML/CSS web interface.
- EduHub** **Team Size: 5**
Frontend Developer **3 Months**
Technologies Used: React, Tailwind CSS, Axios, Node.js, Firebase, AWS S3
Architected a full-stack, multi-tier online coding education marketplace featuring distinct user roles for students, mentors, and administrators. Designed a structured onboarding workflow that allows industry professionals to register as mentors and upload comprehensive programming curricula (covering frontend, backend, cloud, and cybersecurity), which are routed through a secure administrative panel for quality assurance and approval. Implemented a seamless enrollment and tracking engine for students to

discover verified courses, manage their active classes, and programmatically monitor their learning progress through a dynamic user dashboard.

COURSES AND CERTIFICATIONS

- Certified in **Unity Game Development** by **Unity Technologies**, skilled in **C# scripting, gameplay mechanics, and interface design** for **taking interactive 2D and 3D games from raw prototype to a finished experience**.
- Developed expertise in **Frontend Web Development** through a course and internship offered by **IBM**.
- Successfully completed the **Japanese language course** offered by **Cursa**.

ACHIEVEMENTS AND ACTIVITIES

- Worked as a **Design team** member at **GDC MEC 2024**.
- Gained hands-on exposure to **Cyber Security And Ethical Hacking** through a workshop conducted by **Techmaghi** and **IIT Delhi**.
- Participated in **Zilckathon-HFT** a hackathon conducted by **EXCEL MEC 2023**.
- Gained hands-on exposure to **AI with Python** through a training session conducted by **Techmaghi** and **IIT Delhi**.
- **Hobbies:** Chess, Reading, Movies, Gaming

REFERENCES

- **Prof. Dr. Jayachandran E S**, Principal, Govt. Model Engineering College, Kochi, Email ID: principal@mec.ac.in
- **Prof. Dr. Binu V P**, HOD, Computer Science Engineering, Govt. Model Engineering College, Kochi, Email ID: binuvp@mec.ac.in